Parallelism

1. bit/gate level ✓
2. Instruction level ✓

invisible to programmer

3. Thread level
4. Intra-program level ✓

Requires reprogramming

Flynn Taxonomy

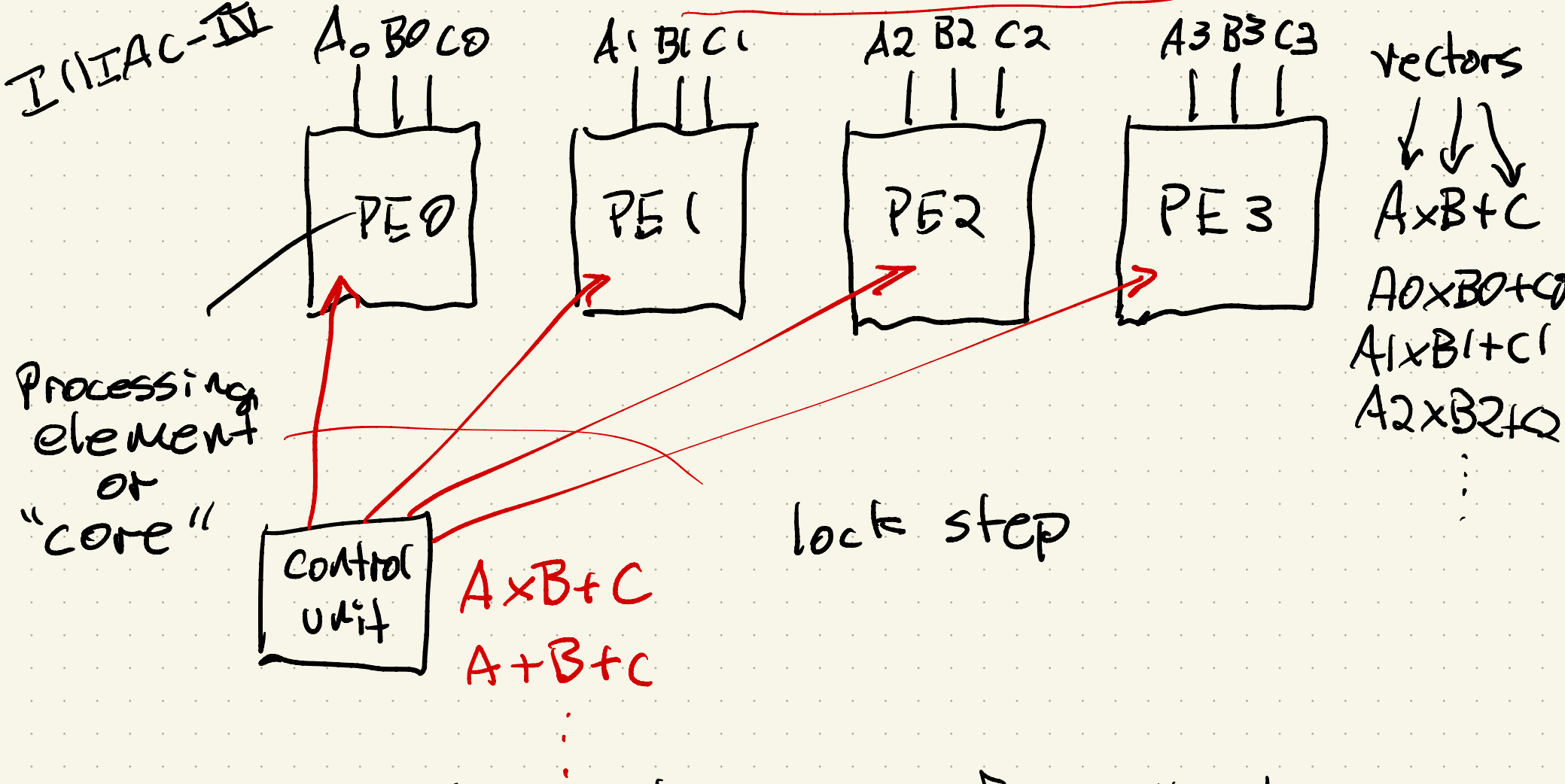
SI SD

MI MD

SISD - simple uniprocessor

SIMD

SIMD is a parallel architecture



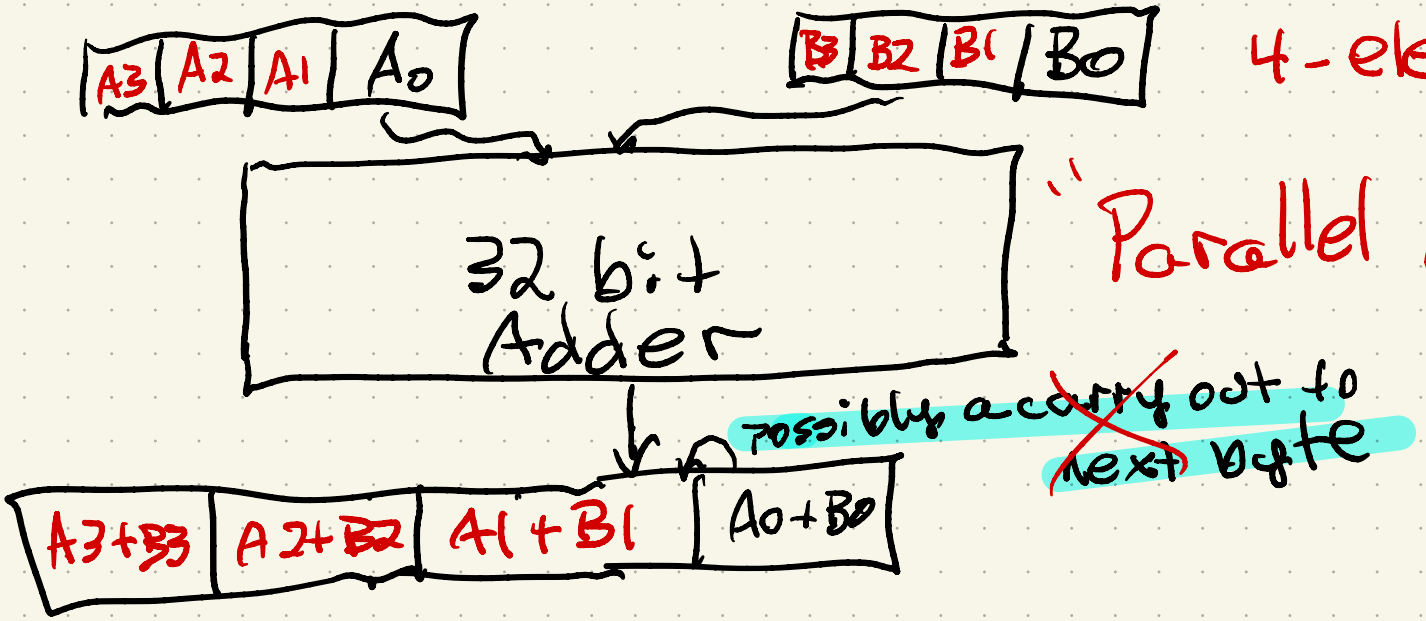
∴ SIMD doesn't work for all algorithms

∵ lock step → all cores are synchronized

bit-level SIMD in uniprocessor

A_i, B_i bytes $A_i + B_i$

registers are 32 bits (4 bytes)

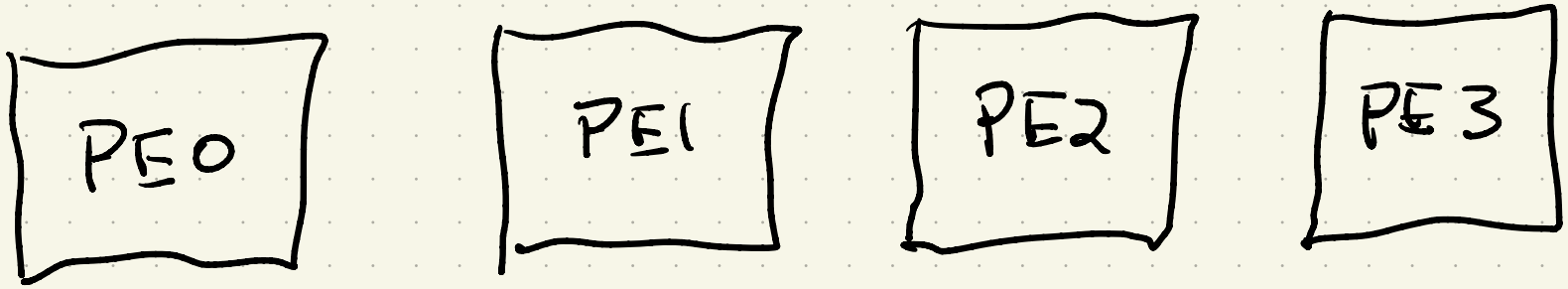


Vector SIMD instructions

"Streaming SIMD instructions"

SSE

MIMD parallel architecture



- Seperate thread per each PE
- No restriction on the inter-thread data use (multiple data streams)

:) All parallel algorithms work on MIMD (includes SIMD algorithms)

:(Threads must synchronize using special synchronization instructions

Hybrids (SIMD on a MIMD)

(Programming models)

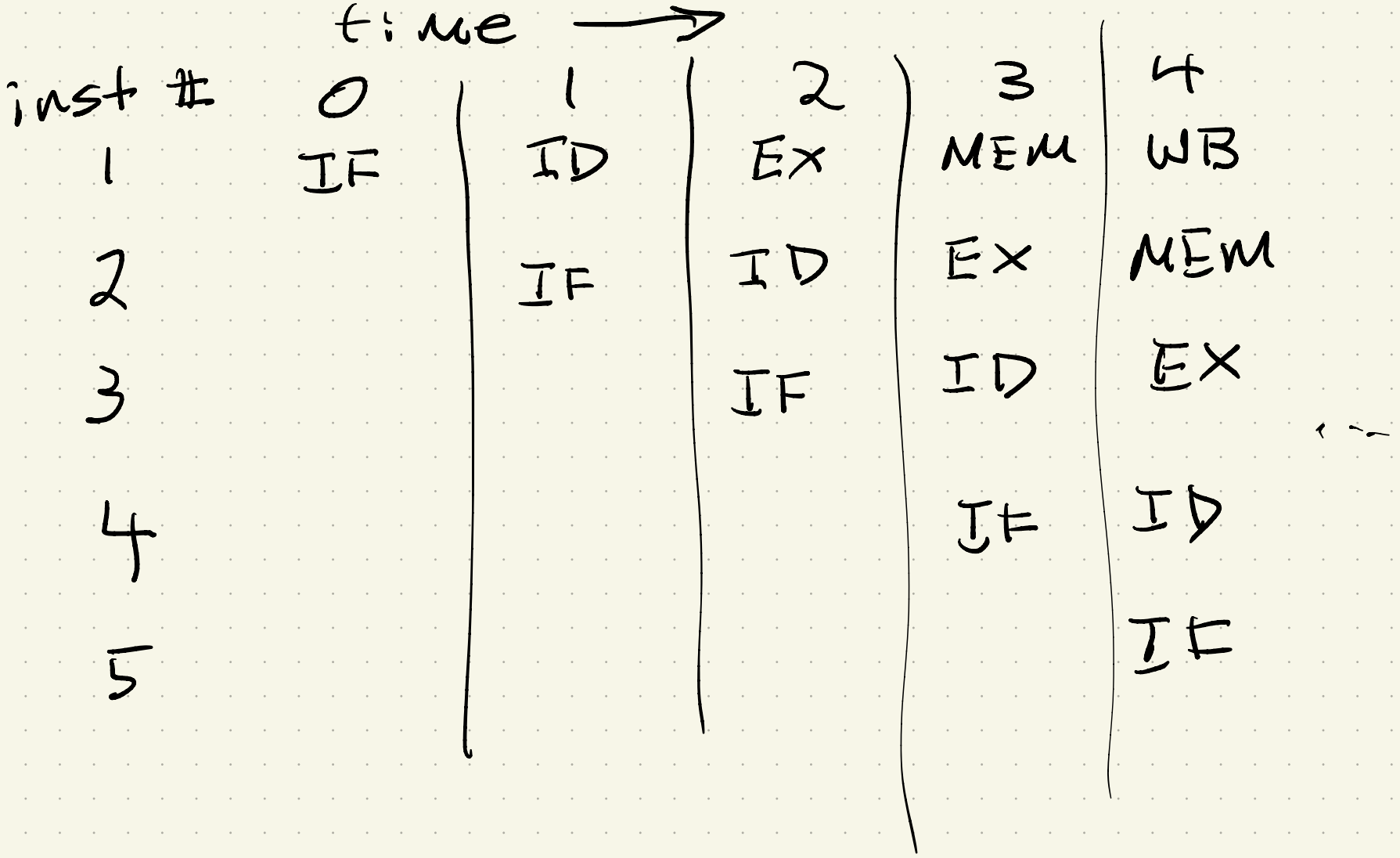
SPMD "Single program multiple data"

All cores execute same thread

```
T1              T2
if (...)        if (...)
  x               x
else            else
  y               y
```

SIMT - same as SPMD

(GPU term for it)



Pipelining - is not SIMD nor MIMD

Vector operations

Floating point add

$1.23 \times 10^3 + 2.34 \times 10^{-1}$

steps

① Denormalize $1230 \quad 0.234$

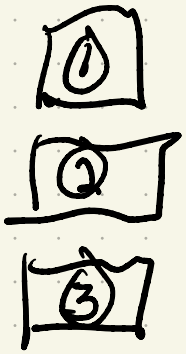
② Add

$1230 + 0.234$

1230.234

③ Renormalize 1.230234×10^3

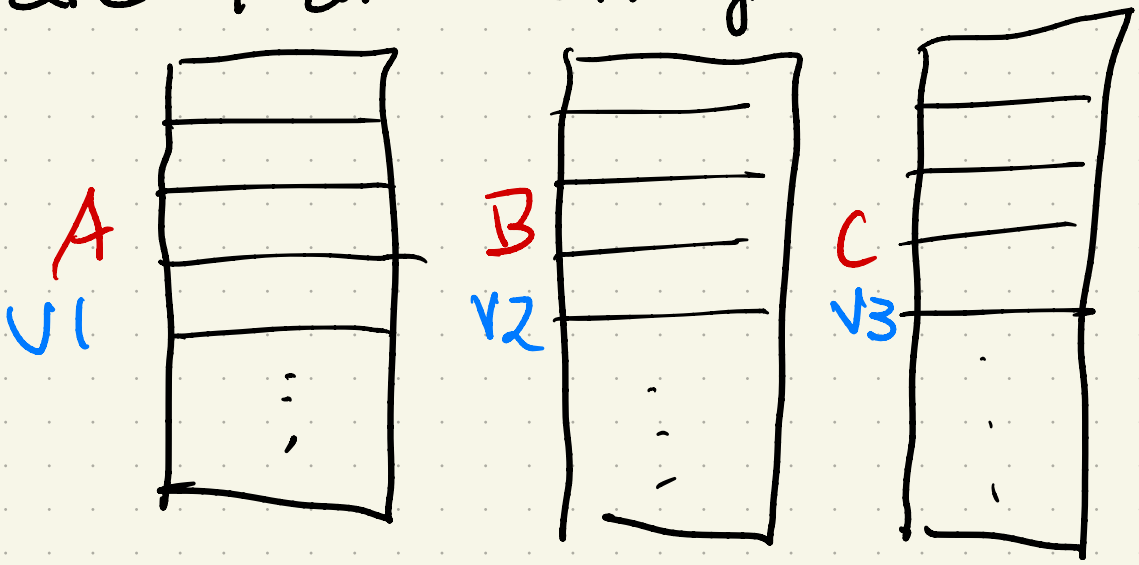
Can break into 3 pipeline stages



Vector execution

Registers are 1-dim. arrays

$\vec{A} \times \vec{B} + \vec{C}$



Seymour Cray

Cray-1

VMUL V4, V1, V2 → 3x faster (3 stages)

VADD V5, V4, V3 → 3x faster (")

"Chaining"

